Database Project Description

Travel Agency

1. Basic project assumptions

The goal of this project is to design a relational database that supports a travel agency organizing guided trips. The system handles registration of people, employees and tourists, planning of routes and trips, ticket sales, payments, group assignment, equipment rental, and service quality ratings.

Main assumptions

- A shared table **Persons** as a base for inheritance for Employees, Guides and Tourists.
- Separation of data that stays the same and data that changes over time.
- Recording the history of guide statuses and the history of route prices.
- Support for many languages for one guide.
- Control of seat availability and correctness of operations.

Limitations

- One person can be only one type of user.
- Limited number of seats on a trip.
- No refund support — every ticket purchase is final.
- Only one active guide status at a given time.

2. Main entities

Main entities: Persons, Employees, Guides, Tourists, Routes, Planned_Trips, Tickets, Trip_Groups, Payments.

Entity attributes

Persons
```
person_id
first_name
last_name
pesel
email
```

Employees
```
employee_id
hire_date
base_salary
```

Guides
```
guide_id
license_number
biography
```

Tourists
```
tourist_id
country_of_origin
passport_number
```

Guide_Status_History
```
history_id
guide_id
status_id
valid_from
valid_to
```

Guide_Status_Dictionary
```
status_id
status_name
```

Routes
```
route_id
route_name
description
difficulty
estimated_time_min
```

Route_Points
```
point_id
route_id
point_name
point_order
```

Price_History
```
price_id
route_id
ticket_type
price
valid_from
valid_to
```

Planned_Trips
trip_id
route_id
guide_id
planned_start
max_people

Trip_Realizations
realization_id
trip_id
actual_start
actual_end
guide_notes

Tickets
ticket_id
tourist_id
trip_id
purchase_date
final_price

Payments
payment_id
ticket_id
payment_method
payment_status

Reviews
review_id
ticket_id
rating
comment

Trip_Groups
group_id
trip_id
group_name

Group_Members
group_id
tourist_id

Guide_Languages
guide_id
language_id
level

Languages
language_id
language_name

Equipment
equipment_id
equipment_name
is_working

Rentals
rental_id
equipment_id
ticket_id
rental_date

3. Database schema

People module

Persons, Employees, Guides, Tourists

Routes and trips module

Routes, Route_Points, Price_History, Planned_Trips, Trip_Realizations

Sales module

Tickets, Payments, Reviews

Groups module

Trip_Groups, Group_Members

Skills and equipment module

Languages, Guide_Languages, Equipment, Rentals

4. Extra integrity constraints

- CHECK on rating in Reviews (1–5)
- UNIQUE on PESEL and license number
- Logic in triggers that blocks incorrect operations

5. Created indexes

This project uses indexes created automatically by the database management system. For every table a primary key index (PRIMARY KEY) was defined. It gives fast access to rows and uniqueness of identifiers. Extra unique indexes were also created for the field **pesel** in table Persons and for **license_number** in table Guides, because of UNIQUE

constraints.

6. Views

v_AvailableGuides

Shows guides who are not currently leading any trip and are available for assignment.

v_SalesReport

Shows a ticket sales report with tourist data, route name, ticket price and purchase date.

v_GroupComposition

Shows the members of trip groups with the route name.

v_GuideSkills

Shows the language skills of guides (languages and skill level).

v_RouteRanking

Shows a ranking of routes based on the number of sold tickets.

v_StatusLogs

Shows the history of guide status changes with validity dates.

v_UpcomingTrips

Shows trips planned for the next 7 days with the number of registered participants.

v_TripsInProgress

Shows trips currently in progress, with guide data and duration from the start time.

v_RentalStatus

Shows the current state of equipment rentals.

v_RouteRevenue

Shows the total revenue from each route based on sold tickets.

7. Stored procedures

sp_ReserveTicket

Reserves a ticket for a tourist on a selected trip. It gets the route id from the trip, uses fn_GetCurrentPrice to get the current price for the ticket type, then inserts a new row into Tickets. This keeps reservation and price calculation consistent.

sp_ChangeGuideStatus

Changes the current status of a guide. First it closes the previous active status by setting valid_to, then it inserts a new row into Guide_Status_History. A guide has only one active status at a given moment.

sp_StartTrip

Starts trip realization. It sets the real start time in Trip_Realizations, then changes the assigned guide status to “On trip” by calling sp_ChangeGuideStatus.

sp_AddRoute

Adds a new route and automatically inserts its first point into Route_Points. Every route has at least one point.

sp_ClosePrice

Archives the current price of a route by setting valid_to for the selected ticket type. This keeps price history and prepares space for a new price.

8. Triggers

tr_ValidateSeats

Runs after a new ticket is added to Tickets. If there are no free seats left, the insert is rolled back.

tr_InitRealization

Runs after a new trip is added to Planned_Trips. It creates a matching row in Trip_Realizations.

tr_BlockPaidTicketDelete

Runs when someone tries to delete a ticket. If the ticket has a payment with status “Paid”, the delete is blocked.

tr_CheckLicense

Runs after insert or update on Guides. It checks the minimum length of the license number.

tr_LogStatusChange

Runs after a new status is added to Guide_Status_History. It closes the previous active status so only one active status exists at a time.

9. Database creation script

The full SQL Server script is in the file **schema.sql** in this repository. It contains table definitions, views, functions, stored procedures, triggers, sample data, and example queries. Run it in Microsoft SQL Server (for example in SQL Server Management Studio or Azure Data Studio).

10. Typical queries

Check free seats

This query checks how many free seats are left on a given trip. It uses function fn_FreeSeats, which subtracts sold tickets from max seats.

```
SELECT dbo.fn_FreeSeats(1) AS free_seats;
```

List of upcoming trips

Shows trips planned for the next 7 days with the number of registered participants.

```
SELECT *
FROM v_UpcomingTrips;
```

Route ranking

Shows a ranking of routes by the number of sold tickets.

```
SELECT *
FROM v_RouteRanking
ORDER BY sold_tickets DESC;
```

Guide language skills

Shows guides with the languages they speak and their skill level. Useful when planning staff for foreign groups.

```
SELECT *
FROM v_GuideSkills;
```

Revenue from routes

Calculates total revenue for each route based on sold tickets.

```
SELECT *
FROM v_RouteRevenue;
```